



Acid-Related Disorders — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which receptor/second-messenger pathway is the final common pathway of parietal cell acid secretion?

EASY

- A. M3 / Gq / IP3
- B. CCK2 / Gq / IP3
- C. H2 / Gs / cAMP
- D. NK1 / Gq / IP3
- E. 5-HT3 / ligand-gated ion channel

2. What is the mechanism of proton pump inhibitors (PPIs)?

EASY

- A. Competitive, reversible H2 receptor blockade
- B. Irreversible, covalent binding to H⁺/K⁺ ATPase
- C. M3 receptor antagonism
- D. Neutralizing gastric acid directly
- E. Inhibiting gastrin release from G cells

3. Which drug is a synthetic prostaglandin E1 analogue used to prevent NSAID-induced ulcers?

EASY

- A. Sucralfate
- B. Misoprostol
- C. Famotidine
- D. Omeprazole
- E. Bismuth subsalicylate

4. What is the standard H. pylori quadruple therapy regimen?

EASY

- A. PPI + amoxicillin + clarithromycin
- B. Bismuth subsalicylate + metronidazole + tetracycline + PPI
- C. H2RA + misoprostol + sucralfate
- D. NaHCO₃ + antacid + antimuscarinic
- E. PPI + azithromycin alone

5. A patient on long-term omeprazole for GERD presents with fatigue and macrocytic anemia. What is the most likely explanation?

MEDIUM

- A. PPI-induced folate deficiency
- B. PPI-induced vitamin B12 deficiency, since gastric acid is needed to release B12 from food
- C. PPI-induced iron deficiency
- D. This is unrelated to the PPI
- E. PPI-induced hemolysis

6. A patient on warfarin is started on cimetidine for dyspepsia and develops a rising INR. What is the mechanism?

MEDIUM

- A. Cimetidine induces CYP450, decreasing warfarin metabolism
- B. Cimetidine inhibits CYP450, potentiating warfarin
- C. Cimetidine displaces warfarin from albumin
- D. Cimetidine increases GI absorption of warfarin
- E. This is unrelated to any drug interaction

7. A patient is prescribed sucralfate for a duodenal ulcer and is also taking an antacid for occasional heartburn, taken at the same time each dose. Why is this combination problematic?

MEDIUM

- A. Antacids and sucralfate cause a disulfiram-like reaction together
- B. Sucralfate requires an acidic pH to activate and form its protective paste; antacids raise gastric pH and reduce its effectiveness
- C. This combination causes fatal hyperkalemia
- D. Sucralfate and antacids compete for the same hepatic enzyme
- E. There is no interaction and this combination is ideal

8. A patient is instructed to take their PPI about 1 hour before a meal rather than with food. What is the pharmacological reason for this timing?

MEDIUM

- A. PPIs are destroyed by any food regardless of timing
- B. Food reduces PPI bioavailability by about 50%, and the drug needs to be active when the proton pump is stimulated by the anticipated meal
- C. PPIs have no relationship to meal timing
- D. Taking with food increases the risk of a disulfiram-like reaction
- E. PPIs require an empty stomach for 12 hours to work

9. A patient with renal failure develops encephalopathy and osteoporosis after chronic use of an aluminum-containing antacid. What is the mechanism?

MEDIUM

- A. Aluminum causes direct hepatotoxicity
- B. Aluminum hydroxide binds phosphate causing hypophosphatemia, and toxicity (encephalopathy, osteoporosis) accumulates specifically in renal failure due to impaired clearance
- C. This is an allergic reaction unrelated to renal function
- D. Aluminum hydroxide causes systemic alkalosis
- E. This is a disulfiram-like reaction

10. A patient takes an antacid at the same time as their daily levothyroxine and ciprofloxacin doses. What is the expected pharmacokinetic consequence?

MEDIUM

- A. No interaction — antacids only affect gastric pH, not drug absorption
- B. Antacids can chelate quinolones like ciprofloxacin and alter the bioavailability of other drugs like levothyroxine, so doses should be spaced apart
- C. Antacids only interact with antibiotics, not thyroid medications
- D. This combination causes acute renal failure
- E. Antacids increase absorption of both drugs, requiring a dose reduction

11. A patient with Zollinger-Ellison syndrome (gastrin-secreting tumor) requires refractory acid suppression despite maximal PPI dosing. An antimuscarinic (e.g. pirenzepine) is added. What is the rationale for this add-on given the three-receptor model of acid secretion?

MEDIUM

- A. Antimuscarinics work by directly inhibiting the H⁺/K⁺ ATPase, the same as PPIs
- B. Antimuscarinics block the M3-mediated (ACh/vagal) component of acid stimulation, providing an additional mechanism distinct from PPI's pump-level blockade
- C. Antimuscarinics have no relevant mechanism for acid secretion
- D. Antimuscarinics work by blocking CCK2 receptors specifically
- E. This add-on is used only to treat nausea, not acid secretion

12. A patient is prescribed misoprostol alongside an NSAID for arthritis to prevent ulcers, but the patient is of childbearing age and sexually active without reliable contraception. What is the key safety concern?

MEDIUM

- A. Misoprostol has no reproductive safety concerns
- B. Misoprostol is contraindicated in pregnancy because it can cause uterine contractions
- C. Misoprostol causes infertility
- D. Misoprostol is only unsafe in postmenopausal women
- E. This concern applies only to sucralfate, not misoprostol

13. A patient with a duodenal ulcer confirmed to be *H. pylori* positive is started on triple therapy. Which combination matches the described first-line regimen?

MEDIUM

- A. Bismuth subsalicylate + metronidazole + tetracycline + PPI
- B. PPI + (metronidazole or amoxicillin) + clarithromycin
- C. H2RA + sucralfate + misoprostol
- D. NaHCO₃ + antimuscarinic + antacid
- E. PPI + ciprofloxacin + doxycycline

14. A patient with congestive heart failure requests an antacid for occasional heartburn. Which antacid type should be specifically avoided given their cardiac condition?

MEDIUM

- A. Aluminum hydroxide alone
- B. Sodium bicarbonate, due to systemic absorption and sodium load which can worsen CHF/edema
- C. Magnesium hydroxide alone
- D. Calcium carbonate is the only safe option regardless of dose
- E. All antacids are equally safe in CHF

15. A patient is switched from an H2 antagonist to a PPI for inadequate acid suppression. The physiology described explains this as a difference in "how" acid secretion is blocked rather than simply potency. Which explanation correctly captures the mechanistic distinction?

HARD

- A. Both drug classes work identically at the molecular level; the difference is purely dose-related
- B. PPIs irreversibly inactivate the H⁺/K⁺ ATPase itself (requiring ~18h to resynthesize new pump), producing longer and more complete suppression, while H2 blockers only competitively and reversibly block the histamine arm, leaving the ACh and gastrin pathways free to still stimulate the pump
- C. H2 blockers work downstream of the pump, while PPIs work upstream at the receptor level
- D. PPIs and H2 blockers both act reversibly, but PPIs simply have a longer half-life
- E. The mechanistic difference is irrelevant to clinical potency

16. A researcher explains that although ACh and gastrin act through Gq/PLC/Ca²⁺-IP3 pathways rather than the Gs/cAMP pathway of histamine, blocking H2 receptors alone still substantially reduces total acid output. What is the best physiological explanation for this, based on the lecture's description of the three-receptor convergence?

HARD

- A. ACh and gastrin have no real effect on acid secretion despite using different pathways
- B. ACh and gastrin partly act by triggering ECL-cell histamine release, so blocking H2 receptors also blunts a component of the ACh/gastrin-driven secretion that operates through this histamine-dependent amplification, even though ACh and gastrin also have some direct parietal cell action
- C. H2 blockade has no measurable effect on total acid output
- D. ACh and gastrin are entirely histamine-independent and this observation would be impossible under the described physiology
- E. Gq and Gs pathways converge identically at the same intracellular step, making them functionally redundant

17. A patient with a history of both peptic ulcer disease and recurrent DVT is being considered for either sucralfate or misoprostol for ulcer prevention while continuing necessary NSAID therapy. Which factor from the lecture's described mechanisms is most relevant to this specific patient's risk-benefit analysis, independent of the pregnancy-related contraindication?

HARD

- A. Sucralfate and misoprostol have identical systemic absorption and cardiovascular risk profiles
- B. Neither drug has any described mechanism relevant to thrombotic risk; the DVT history is not directly addressed by either drug's described mechanism in this lecture, so this factor alone would not differentiate the choice based on the provided content
- C. Misoprostol is specifically contraindicated in any patient with DVT history
- D. Sucralfate is absolutely contraindicated in any patient with a hypercoagulable history
- E. Both drugs directly treat DVT as a secondary indication

18. A patient develops disproportionate diarrhea and loose stools after starting a magnesium-containing antacid, while a different patient develops constipation on an aluminum-containing antacid. A clinician then prescribes a combination $\text{Mg(OH)}_2/\text{Al(OH)}_3$ product. What is the pharmacological rationale for this specific combination?

HARD

- A. The combination has no rationale and simply combines two unrelated side effects
- B. The bowel effects of magnesium (diarrhea) and aluminum (constipation) tend to cancel each other out when combined, as explicitly described
- C. The combination eliminates all antacid-related side effects entirely, including cardiac risk
- D. Combining the two components doubles the antacid potency without any change in side effect profile
- E. This combination is specifically contraindicated and should never be used

19. A patient on chronic PPI therapy for severe GERD is noted to have a theoretically increased infection risk (e.g., *C. difficile*, pneumonia) described in broader acid-suppression literature, though this isn't explicitly detailed in this lecture's ADR list for PPIs. Using only the mechanistic principles given in this lecture (acid's role, e.g., in bacterial killing described for gastric HCl elsewhere in the GIT curriculum), what is the most defensible physiological rationale a student could construct for such a risk, while still being careful not to overstate what this specific lecture states?

HARD

- A. Gastric acid contributes to killing ingested bacteria; profound acid suppression could theoretically reduce this defense, providing a plausible mechanistic rationale for increased enteric infection risk — though the student should note this specific point is inferred from general acid physiology rather than stated explicitly in this pharmacology lecture's ADR list
- B. This risk has no plausible mechanistic basis under any circumstances
- C. PPIs directly cause *C. difficile* infection through a documented cytotoxic mechanism stated explicitly in this lecture
- D. Acid suppression has been proven in this lecture to have zero effect on any infection risk
- E. This reasoning is entirely fabricated and no acid-related mechanism could plausibly explain such a risk

20. A hospital pharmacy committee is evaluating whether to preferentially stock ranitidine or famotidine as the formulary H₂ blocker, given ranitidine's described "less used now — safety concerns" status. What should this status most directly prompt the committee to consider?

HARD

- A. Ranitidine and famotidine are pharmacologically identical, making the choice irrelevant
- B. Given the lecture's explicit note about ranitidine's reduced use due to safety concerns, the committee should weigh this history in favor of preferring famotidine or nizatidine as formulary options over ranitidine, all else being equal
- C. Famotidine should be removed from the formulary instead, since ranitidine is always superior
- D. This status is irrelevant to formulary decisions and should be disregarded
- E. Only cimetidine should be considered, since it has no described interaction profile

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	The H ₂ receptor via Gs/cAMP/PKA is described as the final common pathway of acid secretion.
2	B	EASY	PPIs are prodrugs that irreversibly, covalently bind the H ⁺ /K ⁺ ATPase proton pump.
3	B	EASY	Misoprostol is a synthetic PGE ₁ analogue used for NSAID-ulcer prevention.
4	B	EASY	Quadruple therapy = bismuth subsalicylate + metronidazole + tetracycline + PPI.
5	B	MEDIUM	Acid is required to release B ₁₂ from food; long-term acid suppression can cause B ₁₂ deficiency.
6	B	MEDIUM	Cimetidine inhibits CYP450 (affecting warfarin, phenytoin, theophylline, etc.), potentiating warfarin's effect.
7	B	MEDIUM	Sucralfate needs an acidic environment to activate; combining with antacids (or H ₂ RAs) reduces its efficacy.
8	B	MEDIUM	PPI bioavailability is reduced about 50% by food; timing 1h before a meal also aligns activation with pump stimulation.
9	B	MEDIUM	Al(OH) ₃ binds phosphate (causing hypophosphatemia) and its toxicity (encephalopathy, osteoporosis) is a specific risk in renal failure due to accumulation.
10	B	MEDIUM	Antacid ions chelate quinolones and tetracyclines, and antacids alter bioavailability of several drugs including thyroid hormone; doses should be spaced apart.

11	B	MEDIUM	Antimuscarinics block M3-mediated ACh-driven acid stimulation as an additional mechanism, used as an add-on in refractory PUD/Zollinger-Ellison.
12	B	MEDIUM	Misoprostol is contraindicated in pregnancy due to its uterotonic (uterine contraction-stimulating) effect.
13	B	MEDIUM	Triple therapy is PPI + (metronidazole or amoxicillin) + clarithromycin.
14	B	MEDIUM	NaHCO ₃ is absorbed systemically and carries a sodium load that can worsen CHF/edema; it also risks systemic alkalosis.
15	B	HARD	PPIs irreversibly inactivate the proton pump itself (bottleneck at the final common effector), giving more complete and longer suppression, while H ₂ blockers only competitively block one of three converging inputs (histamine), leaving ACh (M3) and gastrin (CCK2) pathways still able to stimulate acid secretion via the same pump.
16	B	HARD	The lecture notes that ACh and gastrin "also act partly by triggering ECL-cell histamine release" — meaning part of their acid-stimulating effect is histamine-mediated, so H ₂ blockade blunts this shared amplification pathway even though ACh/gastrin also act somewhat directly on the parietal cell via Gq pathways.
17	B	HARD	Neither sucralfate's local mucosal-protective mechanism nor misoprostol's PGE ₁ -analogue mechanism (contraindicated specifically for pregnancy/uterine effects) is described as having a direct thrombotic risk association in this lecture — a careful reader should recognize the DVT history is not addressed by the material provided, rather than assuming an interaction that isn't described.
18	B	HARD	The lecture explicitly notes that combined Mg(OH) ₂ + Al(OH) ₃ antacids have bowel effects that "cancel out" — magnesium's diarrhea tendency offsetting aluminum's constipation tendency.
19	A	HARD	Gastric acid's bactericidal role (documented in general GI physiology) provides a plausible mechanistic bridge to a theoretical infection-risk rationale for profound acid suppression, while appropriately flagging that this specific inference extends beyond what is explicitly stated in this particular lecture's ADR list for PPIs.
20	B	HARD	The lecture explicitly flags ranitidine as "less used now — safety concerns," which is directly relevant information a formulary committee should weigh toward preferring famotidine or nizatidine, all else being equal.

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Begard O. Berzinji's acid-related disorders lecture.
Practice questions — cross-check every answer against the lecture and summary.